“Nestedness in cultural data: a comparative and methodological reassessment”

1. Introduction

We undertake a general overview of nestedness in a broad range of cultural datasets. The notion of nestedness corresponds to a phenomenon whereby assemblages contain both common and rare elements, but tend not to possess exclusively rare elements. Initially introduced in ecology (Ulrich et al., 2009), it is based on the observation that ecosystems rich in rare species also contain common species, while environments with lower biodiversity contain mostly common species. For instance, the set of species in an island usually corresponds to a subset of species from the nearby mainland—a subset composed mostly of common species.

Nestedness is frequently found in domains other than ecology. For example, individuals that know rare facts are also shown to possess extensive general knowledge, but not vice-versa (Dubourg et al., 2025). Additionally, Morin and Sobchuk (2021) have shown that film collections follow a similar pattern: films present in smaller collections are also present in larger collections, but the inverse is rarely true.

Several papers in the literature (e.g. Kamilar & Atkinson, Morin & Sobchuk, Dubourg et al.) argue that cultural datasets are significantly nested across a broad range of domains, from technology to the arts and language. However, this literature suffers from a lack of standardisation of nestedness measurements. Nestedness is a loose concept that can be measured in a variety of ways, and the literature, mostly in ecology, has proposed several metrics. One peculiar feature of these metrics is the fact that they do not necessarily make sense on their own, but must instead be compared to a baseline. Here again, there is no standard in the cultural evolution literature on which baseline to use. Furthermore, somewhat accidental aspects of datasets such as the dimension and fill of the binary matrices on which nestedness is computed are likely to affect the assessment of nestedness.

As a result of this heterogeneity of statistical methods, claims concerning nestedness in cultural evolution are difficult to evaluate and compare. Our goal in this study is to assess nestedness uniformly, applying the same toolkit of metrics and baselines to a series of datasets that have either been analysed and claimed to be nested, or could plausibly be nested given their properties.

Given the relatively small size of the literature on cultural nestedness, we did not undertake a systematic survey of the field, but this study gathers, to our knowledge, all the papers that directly address the issue of cultural nestedness, alongside with some that do not but present relevant features. Our approach is exploratory and not guided by theory-based predictions,

and its focus is methodological. Based on previous work, we believe that assessments of nestedness are keenly sensitive to metrics and baseline, and we wish to observe whether and to what extent such changes can affect nestedness. At the same time, we believe, based on previous work, that nestedness is a potentially important aspect of any dataset shaped by cultural transmission, and we expect to see significant nestedness in the datasets we consider under some metrics and against some baselines.

Our key questions are:

- Is nestedness a pervasive feature of cultural datasets and what are the baselines against which real datasets appear to be significantly nested?
- Does matrix size (a matrix's dimensions) or matrix fill (the proportion of 1 values in the binary matrix) influence nestedness?
- Does nestedness vary from one type of dataset to another (e.g. movie collections, comparative linguistics, trivia questions, etc.)?
- To what extent do our two metrics of choice, NODF and temp, yield overlapping results?

2. Data & methods

2.1. Data sources and pre-processing

In ecology, nestedness analysis often deals with matrices of the following shape : columns are sites and rows are species. The matrix is filled with a binary measure of presence/absence of one species in one site (formally analogous to a bipartite network). The cultural datasets we will study follow the same structure but with cultural repertoires as rows and cultural types as columns.

What we call a repertoire in the context of this study (also sometimes called "collection" or "inventory" in other papers) is a set of cultural items sharing a common origin. For instance, past studies treated the set of movies seen by an agent on Netflix as the agent's movie repertoire (or collection). We also consider as repertoires collections of cultural items such as the inventories of phonemes in languages (each language's inventory being one repertoire), pottery types inventories found in one given archaeological assemblage etc.

What we call a cultural type is something that can be instantiated in several exemplars in various collections. For instance the movie *Fight Club* can be present in several individual collections; the phoneme /a/ may or may not be present in a given phonological repertoire.
In concrete terms, every matrix is organised as follows: the columns (x axis) represent all the types appearing in at least one repertoire, while the rows (y axis) correspond to repertoires (for instance, movie collections or phoneme inventories). Such a binary matrix is identical to a bipartite graph, although not represented with edges and vertices.

To carry out a global analysis of cultural nestedness, we compiled a selection of studies that either directly examine the concept or provide relevant data suitable for such an analysis. In total, our analysis will be based on five research studies, and five additional datasets from the folio R package (Bocquet-Appel & Dubouloz, 2004). The selected articles and datasets are presented below.

The Kamilar & Atkinson (2014) study on cultural assemblages examined six datasets documenting the presence or absence of cultural traits in humans and great apes. For humans, three datasets were used:

- In the Welsch et al. (1992) dataset, each row corresponds to one of 31 villages along New Guinea's north coast, and each column represents one of 47 cultural object classes, such as tools or ornaments; a cell indicates whether a given trait is present in that village.
- The Jordan & Shennan (2003) dataset includes two matrices: one for 43 basketry types and uses, and one for 22 ornamentation traits, with rows representing 39 Californian Indigenous communities.
- In the Jorgensen dataset (1980), 172 tribal sites from western North America form the rows, while 43 technological traits (e.g., clothing, housing) define the columns. Cells indicate presence or absence, with unknowns treated as absences.

For non-human primates, two datasets were used:

- The chimpanzee dataset by Whiten et al. (2001) includes rows for 9 communities across 8 sites and columns for 39 behavioral traits, with three coding variants to handle unknowns: unknowns coded as present, unknowns coded as absent and only habitual and customary traits considered present; all others as absent.
- The orangutan dataset by van Schaik et al. (2009) follows a similar structure with 8 sites and 35 traits, also using the same three different coding schemes.

They quantified nestedness using two standard ecological metrics: NODF (Almeida-Neto et al, 2008) and the temperature metric (Atmar & Patterson, 1993). For each dataset, they assessed significance against randomized baselines using two null models: a fully equiprobable model (ER), which randomizes trait presence without preserving row or column totals, and a more constrained model (CE), which preserves both row and column frequencies. Their results showed highly significant nestedness for all three human datasets and for chimpanzees, across both metrics and baselines. In contrast, none of the three orangutan matrices displayed significant nestedness under any combination of metric or baseline.

Another study that will be included in our analysis is Dubourg et al. (2025), which investigates the nestedness of human knowledge across three domains: superheroes, American history, and astronomy. Each dataset comprises the answers of participants (rows) to 15 domain-specific multiple-choice questions (columns), with correct answers coded as 1

and incorrect as 0. The authors argue that knowledge is nested: individuals who possess rare pieces of knowledge are also likely to possess common ones, while the reverse is not necessarily true. Using standard ecological metrics (NODF and nestedness temperature), they confirmed significant nestedness across all three knowledge domains, using the vegan package's r1 permutation technique as the baseline.

Another study is Morin & Sobchuk (2021), which examines nestedness in user-generated film collections to explain the cultural success of certain movies. The authors propose the "shortlist effect": smaller collections tend to be more selective, favoring highly appealing and widely liked items, which results in stronger nestedness. Their analysis draws on two large film datasets (MovieLens and Netflix) organized by genre. In each dataset, rows correspond to individual user collections and columns to specific films, with cells coded as 1 if a film was rated by a user. Using the NODF metric and a row-fixed, column-proportional baseline (r1), they found significant nestedness in most genres.

Each matrix in this study contains over 5 million cells (see Appendix) and the matrices are extremely sparse. Therefore, due to computational constraints, we worked with subsets of the full matrices. This approach follows the methodology of the referenced study, where a similar subsetting was applied but only to the drama genre matrix. The authors justified their decision by arguing that focusing on widespread films is more meaningful, as these films reach a larger audience and thus better represent the overall film ecosystem. Additionally, such films tend to receive more ratings, making the data less susceptible to randomness. Based on this logic, we retained only the top 10% of the largest collections and the 70% most widely viewed films.

We will also use the phonological inventory dataset covering 1,938 languages from the PHOIBLE 2.0 database (Moran et al., 2019), as utilized in a study by Koshevoy, Morin, and Rebours (in prep.). These languages are grouped into 70 language families, each forming a separate binary matrix in which rows represent individual languages and columns represent phonemes (IPA symbols). Each cell indicates the presence (1) or absence (0) of a given phoneme in a language's inventory. This study explores whether phonological systems exhibit nestedness, and if languages with rare phonemes also tend to include more common ones. The hypothesis is grounded in linguistic theories such as markedness (Haspelmath, 2006) and optimal dispersion (Vaux and Samuels, 2015; Dunbar and Dupoux, 2016), which suggest that phoneme inventories are shaped by systematic principles. Nestedness will be measured for each family matrix.

Another dataset in our sample comes from a study by Salali et al. (2020), which investigated medicinal plant knowledge among BaYaka hunter-gatherers in the Republic of Congo. The dataset includes 219 individuals from four camps (two urban camps and two located in the forest), each represented as a row in a binary matrix. Columns correspond to 33 wild plant species used in traditional medicine, with a value of 1 indicating whether a participant knows or uses each plant. While the original study focused on knowledge retention and the influence

of ecological context on plant knowledge, it did not conduct a formal analysis of nestedness. Nonetheless, the structure of the data, as individual knowledge profiles across a shared set of cultural items, makes it well suited for a comparative analysis of cultural nestedness. The authors of Dubourg et al. conducted a reanalysis of this dataset and found the data to be significantly nested against the r1 baseline, using the NODF metric (Dubourg personal communication).

In addition to these studies, we included seven archaeological assemblage datasets derived from the folio R package (Bocquet-Appel & Dubouloz, 2004), which documents the presence of ceramic or lithic types across various temporal or spatial contexts. These matrices are well-suited to nestedness analysis due to their structured inventory data shape. The *Boves* (Racinet, 2002) dataset includes 8 chronological phases (rows) and 13 ceramic types (columns). The *Chevelon* (Reid, 1982) dataset consists of 12 archaeological sites and 10 ground stone types. *Compiegne* (Lacroix, 1997) includes 5 periods and 16 ceramic types. The *Merzbach* (Crema et al., 2016) dataset features 8 chronological phases and 36 pottery motifs. Lastly, the *Loire* (Husi, 2022) assemblages are represented in three distinct matrices, grouping data by sites (88 rows), cities (54 rows), and areas (12 rows), each with 326 ceramic types. All matrices were binarized by coding any occurrence count greater than 0 as presence (1), and 0 as absence.

Overall, our analysis is based on a total of 122 binary matrices drawn from a diverse set of cultural domains. These include:

- 7 matrices from archaeological pottery assemblages from the R package folio;
- 10 matrices from Kamilar and Atkinson's (2014) comparative study of cultural traits in chimpanzees and orangutans;
- 70 phonological matrices from the PHOIBLE database (comprising 70 language families);
- 3 matrices from Dubourg et al.'s study (2025) on trivia knowledge across different domains (superheroes, astronomy, and American history);
- 2 matrices from Salali et al.'s study (2020) of medicinal plant knowledge among the BaYaka;
- 29 matrices from Morin and Sobchuk's analysis of user movie collections (15 from MovieLens and 14 from Netflix, each broken down by genre).

These 122 together provide a broad sample of cultural data suitable for comparative nestedness analysis. All matrices are listed in the appendix, see figure 1 for an overview of matrices.

Article	Rows (repertoires)	Columns (cultural types)	Cell Values	Metrics used	Baselines used
Kamilar & Atkinson (Cultural Assemblages, 2014)	Cultural groups (villages, tribes, primate populations) and their technologies	Cultural traits or object types	1 = present; 0 = absent	NODF + Temperature	ER (Equiprobable/Random) + CE (Column and Row Totals Equalized)
Dubourg et al. (Nested structure of knowledge, 2025)	Participants and their answers to trivia questions	15 questions per domain (Superheroes, history, astronomy)	1 = correct answer; 0 = incorrect answer	NODF + Temperature	r1 vegan
Morin & Sobchuk (Shortlist Effect, 2021)	Users of the Netflix and MovieLens datasets and their movie collections	Movies	1 = movie rated; 0 = not rated	NODF	r1 vegan + synthetic simulations
Koshevoy, Morin & Rebours (Phonological Inventories, 2025)	Individual languages and their phoneme inventories	Phonemes (IPA symbols)	1 = phoneme present; 0 = absent	NODF + Temperature	r1 + c0 (vegan)
Salali et al. (Global WEIRDing, 2020)	Individual BaYaka participants and their knowledge of plants	Wild plant species	1 = plant known or used; 0 = unknown or unused	No nestedness analysis	No nestedness analysis
Archeology Pots (Bocquet-Appel & Dubouloz, 2004)	Temporal or spatial context and related archaeological finds	Ceramic and lithic types	Depends	No nestedness analysis	No nestedness analysis

Figure 1. Summary of datasets and methods in nestedness research across Cultural domains

Each matrix was extracted directly from the datasets provided by the corresponding articles or datasets. When necessary, the data was binarized to fit the presence–absence structure required for nestedness analysis. For instance, in the archaeological datasets from the folio R package, the original values represented counts. These were transformed into binary format: all values greater than 0 were recoded as "1", while values equal to 0 remained as "0".

In the matrices we analyse, it is often the case that “0” values cannot be straightforwardly interpreted as the sheer absence of a cultural type from a repertoire, since data loss in many forms can intervene. For instance, in the Netflix movie datasets a person may see a movie outside the platform but not on Netflix, in which case the movie would be marked as “0” for their repertoire. Likewise, archaeological datasets are prey to data loss and taphonomic biases (Perreault 2019). In our interpretation, “1” values are best seen as an indication that a cultural type has exceeded a certain threshold for popularity, frequency, or exposure, in a certain repertoire, and a “0” value, that the cultural type has not been important enough, rather than 0s standing for absence and 1s for presence.

In the plant knowledge dataset from Salali et al. (2020), one cell contained a missing value (NA). To account for this, we generated two versions of the matrix: one where the NA was treated as a "1" (indicating knowledge of the plant), and one where it was treated as a "0" (indicating no knowledge). This decision is based on the nature of the original survey, in which participants were asked whether they knew and used specific plants. Since the only possible responses were binary, either knowledge (1) or no knowledge (0), we will consider both interpretations of the missing value to explore its impact on the nestedness results.

All matrices were subsequently cleaned to ensure a consistent and uniform format across the dataset. This cleaning process involved removing all row and column labels, leaving only the binary data (0s and 1s). By standardizing the structure in this way, we ensured that each matrix was fully compatible with the nestedness analysis pipeline and could be processed automatically without additional formatting adjustments.

2.2. Data analysis

To quantify the degree of nestedness, we use two measures derived from ecology: the Nestedness Temperature Calculator (NTC) and the Nestedness metric based on Overlap and Decreasing Fill (NODF), both available in the vegan package in R (Oksanen et al., 2025).

The NTC, proposed by Atmar & Patterson (1993) (see also a review by Rodríguez-Gironés & Santamaria, 2006) measures the degree of randomness in a matrix by comparing its organisation with a theoretically perfect order. It is based on the concept of surprisal, which quantifies the unpredictability of presence and absence. A high temperature indicates high

variability and low nestedness, while a low temperature reflects a more systematic, non-random, structure. This measure is standardized on a scale from 0, which corresponds to perfect nestedness to 100, corresponding to lowest nestedness.

The NODF measure, introduced by Almeida-Neto et al (2008), measures the nestedness of a matrix by evaluating two criteria: the progressive decrease in marginal totals, where the biggest inventories must contain more 1s than the least rich, and pair overlap, which checks whether the items of the least rich repertories are systematically a subset of those of the richest repertories. This measure is also normalized between 0 (maximum disorder) and 100 (perfect nestedness).

Once we will compute these measures for every matrix in our sample, we will compare these with a random baseline, to determine whether the observed degree of nestedness is not due to chance. To do this, we will generate simulated matrices using eight null models: r00, r0, r1, r2, c0, c1, curveball and swap.

Below is a description of the baselines we will use in the the vegan package in R :

- r00: This is a non-sequential algorithm that preserves only the total number of presences (i.e., the number of 1s in the matrix). It does not maintain the original row or column totals, making it the most relaxed constraint model.

- r0: This non-sequential algorithm maintains the row sums, meaning that the number of presences per agent (rows) is preserved. However, the distribution of presences across items (columns) is allowed to vary.

- r1: Also preserving row sums, this algorithm differs from r0 by introducing a probabilistic element. Items are selected based on their original column frequencies, which are used as probabilities when assigning presences.

- r2: A variation of r1, this method still preserves row totals but places greater emphasis on items that were more frequent in the original matrix. It does so by using the squared column frequencies as selection probabilities.

- c0: This non-sequential algorithm preserves column sums, ensuring that item frequencies (i.e., the number of agents each item appears in) remain the same. It follows the approach introduced by Jonsson (2001).

- swap: Unlike the others, this is a sequential algorithm. It preserves both row and column sums by performing a series of swaps within 2×2 submatrices where swaps are possible. This approach maintains marginal sums while altering the internal structure of the matrix (Gotelli & Entsminger 2003).

- c1: This baseline does not exist in the vegan package in R. It is the column-wise counterpart to r1. We implement it by flipping the matrix and then applying r1 to this new matrix with rows representing the old columns, and columns the old rows. To clarify, c1 is a non-sequential algorithm preserving column sums, this algorithm differs from c0 by introducing a probabilistic element. Items are selected based on their original row frequencies, which are used as probabilities when assigning presences.

- curveball: sequential method for binary matrices that implements the 'Curveball' algorithm of Strona et al. (2014). The algorithm selects two random rows and finds the set of unique species that occur only in one of these rows. The algorithm distributes the set of unique species to rows preserving the original row frequencies. Zero to several species are swapped in one step, and usually the matrix is perturbed more strongly than in other sequential methods.

Another property of the matrices that we will compute is the coefficient of correlation between a cultural type's prevalence and the average inventory size of the inventories where it is found. That is to say, for each cultural type, we calculate the number of repertoires in which it is found (type prevalence) and we take the mean size of those repertoires (average inventory size of repertoires possessing the cultural type of interest). We then compute Pearson's coefficient of correlation between those two measurements for all cultural types in the matrix. If this coefficient is close to -1, it means that the less likely a cultural type is to appear in repertoires, the bigger the repertoires that possess it tend to be. A negative correlation is implied by cultural nestedness (Dubourg et al. 2025).

2.3. Dataset structure

For each of the 122 matrices in our dataset, nestedness will be assessed using two metrics, NODF (Almeida-Neto et al, 2008) and temperature (Atmar & Patterson, 1993), across eight different null models: r00, r0, r1, r2, c0, c1, curveball, and swap. For each matrix and each baseline, 1,000 random matrices will be generated using the null model function from the vegan R package,and the simulate function.Then we computed nestedness with both NODF and Temp on these 1,000 simulated matrices. This will result in 8,000 simulated values per matrix. For each matrix, we will produce two types of CSV files: (A) a detailed file containing all simulation results (8,000 rows × 6 columns), and (B) a summary file with key statistics of the real matrix (1 row × 10 columns). This structure allows for comprehensive statistical comparisons of observed nestedness values against a range of null models, ensuring consistent evaluation across metrics and datasets.

We will then compute separately a two sided bootstrapped p-value for each matrix, indicating how many of the simulated matrices display a higher or lower NODF score or a higher or lower temperature than the empirical matrix. A high proportion of such occurrences suggests that the structure observed in the empirical matrices is not due to chance.

Structure of Dataset A (Simulations Data):

This file contains the detailed output of the nestedness simulations for each matrix. Each row corresponds to a simulated matrix under a given baseline.

1. ID – Unique identifier for the matrix (e.g. corresponding to a language family, or cultural domain)

2. Baseline – The null model used to generate the simulated matrix (e.g. r00, r1, c0, etc.)

3. stat_NODF_general – The overall NODF score for the matrix, where higher values indicate higher nestedness

4. stat_NODF_col – The NODF score of columns (item overlap)

5. stat_NODF_row – The NODF score of rows (agent overlap)

6. stat_Temp – The nestedness temperature score, where lower values indicate higher nestedness

Structure of Dataset B (Real Matrix Data):

This file contains summary statistics and metadata for each empirical matrix.

1. ID – Unique identifier for the matrix

2. n_row – Number of rows (e.g. number of languages/sites/participants)

3. n_col – Number of columns (e.g. number of phonemes/questions)

4. fill – Matrix density, i.e., proportion of cells filled with 1s

5. size – Total number of cells in the matrix (n_row × n_col)

6. coef_cor – Correlation coefficient between item prevalence (number of agents possessing the item) and average inventory size of agents possessing said item, measuring structural association
7. stat_NODF_general – Empirical NODF score for the matrix

8. stat_NODF_col – Empirical NODF score of columns

9. stat_NODF_row – Empirical row-wise NODF score of rows

10. stat_Temp – Empirical temperature score

Structure of dataset C (oecosimu table)

1. p_value_NODF_[baseline] – p-values from simulations for the general NODF score, reported for each of the 8 baselines

2. p_value_Temp_[baseline] – p-values from simulations for the temperature score, reported for each of the 8 baselines

3. References

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Appendix: List of matrices included in the study ordered by size

Group	File	Rows	Columns	Fill (%)	Size
Movies	ml_drama_filtered.csv	12231	3092	2,02	37 818 252
Movies	ml_comedy_filtered.csv	12304	1576	3,50	19 391 104
Movies	nf_thriller_filtered.csv	43905	419	18,10	18 396 195
Movies	nf_comedy_filtered.csv	38705	351	13,89	13 585 455
Movies	nf_action_filtered.csv	38395	325	12,08	12 478 375
Movies	nf_drama_filtered.csv	35025	296	10,42	10 367 400
Movies	nf_scifi_filtered.csv	37552	259	18,90	9 725 968
Movies	ml_crime_filtered.csv	12042	778	5,00	9 368 676
Movies	nf_adventure_filtered.csv	34892	261	16,12	9 106 812
Movies	nf_family_filtered.csv	37419	241	16,26	9 017 979
Movies	nf_romance_filtered.csv	34165	255	10,40	8 712 075
Movies	nf_musical_filtered.csv	32819	255	10,42	8 368 845
Movies	nf_biopic_filtered.csv	38908	203	15,59	7 898 324
Movies	nf_fantasy_filtered.csv	38312	197	18,22	7 547 464
Movies	nf_horror_filtered.csv	25253	291	12,35	7 348 623
Movies	ml_thriller_filtered.csv	11359	607	4,31	6 894 913
Movies	ml_documentary_filtered.csv	4520	1316	1,45	5 948 320
Movies	ml_romcom_filtered.csv	11392	519	6,06	5 912 448
Movies	nf_historical_filtered.csv	30389	169	11,75	5 135 741
Movies	ml_action_filtered.csv	9124	549	4,34	5 009 076

Movies	ml_war_filtered.csv	10511	439	4,55	4 614 329
Movies	ml_horror_filtered.csv	5781	730	4,32	4 220 130
Movies	ml_scifi_filtered.csv	9757	415	5,50	4 049 155
Movies	ml_fantasy_filtered.csv	9668	282	6,77	2 726 376
Movies	nf_animation_filtered.csv	18919	136	16,39	2 572 984
Movies	ml_adventure_filtered.csv	10989	226	9,86	2 483 514
Movies	ml_western_filtered.csv	6061	315	4,32	1 909 215
Movies	ml_animation_filtered.csv	5952	274	4,96	1 630 848
Movies	ml_noir_filtered.csv	2542	152	7,74	386 384
Phoible	Atlantic_Congo.csv	423	739	5,48	312 597
Phoible	Indo_European.csv	149	773	5,46	115 177
Phoible	Sino_Tibetan.csv	100	550	8,1	55 000
Phoible	Afro_Asiatic.csv	108	469	8,91	50 652
Phoible	Pama_Nyungan.csv	253	166	14,66	41 998
Phoible	Austronesian.csv	114	322	8,12	36 708
Archeology	loire_site.csv	88	326	6,03	28 688
Archeology	loire_city.csv	54	326	6,67	17 604
Phoible	Austroasiatic.csv	40	304	13,65	12 160
Phoible	Uralic.csv	32	350	12,32	11 200
Phoible	Mande.csv	43	176	22,33	7 568
Human	jorgensen_etal1980.csv	172	43	58,17	7 396
Plant Knowledge	plant_knowledge_0.csv	219	33	80,66	7 227
Plant Knowledge	plant_knowledge_1.csv	219	33	80,7	7 227
Phoible	Dravidian.csv	33	208	19,03	6 864

Phoible	Arawakan.csv	43	145	17,96	6 235
Phoible	Nilotic.csv	28	209	19,5	5 852
Phoible	Tupian.csv	50	104	24,33	5 200
Phoible	Turkic.csv	21	244	16,3	5 124
Phoible	Nuclear_Trans_New_Guinea.csv	30	156	14,51	4 680
Trivia	superheroes.csv	287	15	41,35	4 305
Trivia	solarsystem.csv	282	15	45,3	4 230
Trivia	americanhistory.csv	279	15	31,09	4 185
Phoible	Otomanguean.csv	17	239	16,96	4 063
Phoible	Central_Sudanic.csv	22	179	24,71	3 938
Archeology	loire_area.csv	12	326	19,73	3 912
Phoible	Nakh_Daghestanian.csv	13	272	21,83	3 536
Phoible	Tai_Kadai.csv	19	160	22,7	3 040
Phoible	Uto_Aztecan.csv	15	128	21,98	1 920
Phoible	Nuclear_Macro_Je.csv	20	94	30,27	1 880
Phoible	Mongolic.csv	12	155	23,6	1 860
Phoible	Quechuan.csv	23	80	30,33	1 840
Phoible	Pano_Tacanan.csv	24	72	29,51	1 728
Human	JordanShennanTypesUses.csv	39	43	29,1	1 677
Phoible	Athabaskan_Eyak_Tlingit.csv	10	164	27,68	1 640
Phoible	Cariban.csv	24	61	35,11	1 464
Human	welsch_etal1992.csv	31	47	46,67	1 457
Phoible	Salishan.csv	11	129	31,43	1 419
Phoible	Ta_Ne_Omotic.csv	10	135	28,59	1 350
Phoible	Mayan.csv	12	100	31,08	1 200

Phoible	Tucanoan.csv	19	60	36,05	1 140
Phoible	Gunwinyguan.csv	14	71	37,73	994
Phoible	Chibchan.csv	11	79	33,03	869
Human	jordanshennanornamentation.csv	39	22	25,17	858
Phoible	Kadugli_Krongo.csv	7	90	63,97	630
Phoible	Kiowa_Tanoan.csv	5	126	37,94	630
Phoible	Algic.csv	6	86	33,72	516
Phoible	Heibanic.csv	6	85	45,49	510
Phoible	Surmic.csv	5	84	45	420
Phoible	Tungusic.csv	5	80	37,5	400
Phoible	Ijoid.csv	6	62	64,52	372
Phoible	Sepik.csv	7	53	37,74	371
Phoible	Songhay.csv	4	92	52,72	368
Chimpanzee	whiten_habitualcustomary_only.csv	9	39	28,49	351
Chimpanzee	whiten_unknowns_are_absent.csv	9	39	36,75	351
Chimpanzee	whiten_unknowns_are_present.csv	9	39	49,86	351
Phoible	Nyulnyulan.csv	10	34	65,59	340
Phoible	Eskimo_Aleut.csv	5	66	40,61	330
Phoible	Chocoan.csv	6	54	58,02	324
Phoible	Mixe_Zoque.csv	6	54	48,46	324
Phoible	Nuclear_Torricelli.csv	6	54	41,36	324
Phoible	Western_Daly.csv	6	53	51,26	318
Phoible	Kartvelian.csv	4	73	53,08	292

Phoible	Nambiquaran.csv	5	58	52,41	290
Archeology	merzback.csv	8	36	65,28	288
Orangutan	vanschaik2009_habitualcustomary_only.csv	8	35	22,86	280
Orangutan	vanschaik2009_unk_absent.csv	8	35	40,71	280
Orangutan	vanschaik2009_unk_present.csv	8	35	45,71	280
Phoible	Chukotko_Kamchatkan.csv	4	68	37,5	272
Phoible	Guaicuruan.csv	5	51	50,2	255
Phoible	Narrow_Talodi.csv	5	50	58,8	250
Phoible	Worrorran.csv	8	31	81,45	248
Phoible	Mirndi.csv	8	30	69,17	240
Phoible	Arawan.csv	5	46	49,13	230
Phoible	Barbacoan.csv	4	56	43,75	224
Phoible	Matacoan.csv	4	55	49,55	220
Phoible	Nadahup.csv	4	54	63,43	216
Phoible	Saharan.csv	4	47	68,62	188
Phoible	Iroquoian.csv	4	45	41,67	180
Phoible	Guahiboan.csv	4	44	55,68	176
Phoible	Huitotoan.csv	4	43	60,47	172
Phoible	Muskogean.csv	4	41	53,66	164
Phoible	Iwaidjan_Proper.csv	5	30	79,33	150
Phoible	Mangarrayi_Maran.csv	4	37	61,49	148
Phoible	Maningrida.csv	4	37	63,51	148
Phoible	Yanomamic.csv	4	37	72,97	148
Phoible	Zaparoan.csv	4	33	63,64	132
Phoible	Chicham.csv	4	32	73,44	128

Archeology	chevelon.csv	12	10	48,33	120
Phoible	Koman.csv	4	30	88,33	120
Archeology	boves.csv	8	13	94,23	104
Archeology	compiegne.csv	5	16	95	80